

Question Number	Answer	Additional Guidance	Mark
1(a)	A description that contains five of the following: • pondweed set up with suitable apparatus to collect gas (1) • use of at least two filters of {different / named} colours or one (coloured) filter and white light (1) • sodium hydrogen carbonate added (in excess) (1) • give pondweed time to acclimatise (1) • volume of gas collected in stated time (1) • identification and control of one variable (1) • repeats and calculate means (to compare) (1)	Accept diagram of apparatus Accept named wavelengths (instead of colours) eg in the range 480 – 750nm eg 3-30 minutes eg. temperature – eg use a heat shield pH – buffer length of pondweed – measure light intensity – bulb at fixed distance / wattage	(5)

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1(b)	An explanation that includes three of the following: • (grana formed from) stacks of membrane (1) • therefore providing large surface area (for photosystems / enzymes) (1) • containing {pigments / named pigments} to absorb light (1) • containing {enzymes / named enzymes} (for light dependent reactions) (1)	Accept pigments releasing excited electrons Accept containing {electron carriers / electron transport chain} Accept enzymes providing {materials / named materials} for the light independent reactions.	(3)

(Total for Question 1 = 8 marks)

